

Tide retrospective: J-function asymptotic-equivalence wrapper

Daniel Murfet

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Abstract

This is a 200-line tide that closes the J-function programme’s narrative: with the parametric form $t^{1/(2k)} \cdot J_k(t) \rightarrow \pi(0) \cdot \text{CONST}_k$ already established by the generic- k tide (N1) and the matching partition $Z_k(t)$ available in closed form, the asymptotic-equivalence statement actually used in the grammar paper,

$$J_k(t) \sim \pi(0) \cdot Z_k(t) \quad (t \rightarrow \infty),$$

collapses to a thin `Tendsto.div_const` corollary plus a Mathlib idiom-bridge.¹ The unconditional ratio form $J_k/Z_k \rightarrow \pi(0)$ is the cleaner statement and the building block; the *IsEquivalent* form is the named user-facing corollary under $\pi(0) \neq 0$. Single session, GPT-5.5 Pro and Claude voted Candidate B (both theorems) on the first round, zero-friction proof, 0 `sorry`.

1 Setting

The grammar-paper precursor sequence has three landings on 7 May: quartic ($k = 2$, the E4 tide), sextic ($k = 3$, the L1 tide), and generic- k (the N1 tide). Each established the same shape of asymptotic in a $t^{1/(2k)}$ -rescaled form:

$$t^{1/(2k)} \cdot J_k(t) \longrightarrow \pi(0) \cdot \frac{1}{k} \cdot ((2k)!)^{1/(2k)} \cdot \Gamma(1/(2k)),$$

which is mathematically equivalent to (but typographically distinct from) the form the grammar paper actually uses:

$$J_k(t) \sim \pi(0) \cdot Z_k(t),$$

where $Z_k(t) = \int e^{-tw^{2k}/(2k)!} dw$ is the matching partition. The L1 tide explicitly named this conversion as a follow-up: “*The grammar paper’s natural statement is $J(t) \sim \pi(0) \cdot Z(t)$... Conversion is `Tendsto.div` against `quartic_partition`’s positive-and-tending-to-zero $Z(t)$. ~10 lines on top of `JFunctionQuartic.lean`.*”

The L1 estimate of ~10 lines was for the $k = 2$ specialisation. After N1, the right place to land the wrapper is at the *parametric* level — once on `JFunctionMonomial.lean`, valid for all $k \geq 1$, with the $k = 2$ and $k = 3$ specialisations as zero-line corollaries (instantiate k). The actual implementation took ~200 lines, an overrun driven by closed-form-positivity bookkeeping rather than by any mathematical surprise.

This tide is therefore another *narrative-closer*: it does not prove a new asymptotic content; it packages the existing asymptotic into the named idiom the paper uses.

¹Specifically `Asymptotics.isEquivalent_iff_tendsto_one`.

2 The thing formalised

For the single-monomial potential $K(w) = w^{2k}/(2k)!$ with $k \geq 1$ and a continuous, globally bounded prior π , the J -function

$$J_k(t) := \int_{\mathbb{R}} e^{-t w^{2k}/(2k)!} \pi(w) dw$$

and the matching partition

$$Z_k(t) := \int_{\mathbb{R}} e^{-t w^{2k}/(2k)!} dw$$

satisfy two convergence statements as $t \rightarrow \infty$:

1. (Unconditional ratio.) $\frac{J_k(t)}{Z_k(t)} \rightarrow \pi(0)$.
2. (Asymptotic-equivalence.) Under $\pi(0) \neq 0$, $J_k \sim \pi(0) \cdot Z_k$ in the sense that $J_k - \pi(0) Z_k = o(\pi(0) Z_k)$.

The Lean signatures, in `Laplace/OneD/JFunctionAsymptoticEquivalence.lean`²:

```
theorem kth_jfunction_div_partition
  {k : ℕ} (hk : 1 ≤ k)
  {prior : ℝ → ℝ} (hprior_cont : Continuous prior)
  {M : ℝ} (hprior_bd : ∀ x, |prior x| ≤ M) :
  Tendsto (fun t : ℝ =>
    (∫ w, exp(-(t * w^(2k) / (2k)!)) * prior w) /
    (∫ w, exp(-(t * w^(2k) / (2k)!))))
    atTop (ℕ (prior 0))

theorem kth_jfunction_isEquivalent
  {k : ℕ} (hk : 1 ≤ k)
  {prior : ℝ → ℝ} (hprior_cont : Continuous prior)
  {M : ℝ} (hprior_bd : ∀ x, |prior x| ≤ M)
  (hprior0 : prior 0 ≠ 0) :
  (fun t : ℝ => ∫ w, exp(-(t * w^(2k) / (2k)!)) * prior w)
    ~[atTop]
  (fun t : ℝ => prior 0 * ∫ w, exp(-(t * w^(2k) / (2k)!)))
```

The first is the building block; the second is the user-facing form.

3 Proof strategy

The core observation is short: the partition $Z_k(t)$ has an *exact* closed form, not just an asymptotic. From N1's `kth_partition` we get

$$Z_k(t) = \frac{1}{k} ((2k)!/t)^{1/(2k)} \Gamma(1/(2k)) \quad \text{for } t > 0,$$

²The file extends `JFunctionMonomial.lean` (the N1 seabed); it imports `Mathlib.Analysis.Asymptotics.AsymptoticEquivalent` for the named idiom `IsEquivalent`. Permalink: the file at the new commit on the tide branch.

which, with $\alpha := 1/(2k)$ and $\text{CONST}_k := (1/k) \cdot ((2k)!)^\alpha \cdot \Gamma(\alpha)$, rearranges to

$$t^\alpha \cdot Z_k(t) = \text{CONST}_k \quad (\text{exact, for all } t > 0).$$

Combined with N1’s `kth_jfunction_asymptotic`, which gives $t^\alpha \cdot J_k(t) \rightarrow \pi(0) \cdot \text{CONST}_k$, the ratio is

$$\frac{J_k(t)}{Z_k(t)} = \frac{t^\alpha \cdot J_k(t)}{t^\alpha \cdot Z_k(t)} = \frac{t^\alpha \cdot J_k(t)}{\text{CONST}_k} \rightarrow \frac{\pi(0) \cdot \text{CONST}_k}{\text{CONST}_k} = \pi(0).$$

Implementation in Lean: `Tendsto.div_const` on the existing N1 limit, plus a `Tendsto.congr’` to swap the form.

The *IsEquivalent* corollary then uses Mathlib’s `isEquivalent_iff_tendsto_one`: under eventual non-vanishing of the target $\pi(0) \cdot Z_k(t)$, $f \sim g$ is equivalent to $f/g \rightarrow 1$. The eventual-nonvanishing condition reduces to $\pi(0) \neq 0$ (since $Z_k(t) > 0$ for all $t > 0$ by the closed form, which is a positive product of positive factors). The ratio convergence $f/g \rightarrow 1$ is then $(J_k/Z_k)/\pi(0) \rightarrow \pi(0)/\pi(0) = 1$, which follows from the unconditional ratio theorem by a single `Tendsto.div_const`, modulo a `Pi.div` reshape.

The mathematical narrative is unsurprising; the Lean encoding is mechanical. The 200-line file size comes from the closed-form positivity bookkeeping (positivity of the constant, of the `rpow`, of the Gamma function, of $(2k)!/t$ for $t > 0$) and from the constant-massaging `set` block at the top of the unconditional theorem, not from any clever proof.

4 Roadblocks and resolutions

Three small Lean-level frictions, all anticipated by the laplace `CLAUDE.md`.

One transient “no goals” from over-eager `field_simp`. The closed-form-rearrangement $t^\alpha \cdot Z_k(t) = \text{CONST}_k$ chains `rw [kth_partition hk ht]` with `rw [Real.div_rpow hfacnn htann]` and finishes with `field_simp`. The first draft included a trailing `ring` out of habit; `field_simp` had already closed the goal, producing the error “No goals to be solved.” Fix: drop the `ring`. The lesson is the laplace `CLAUDE.md` entry on “`field_simp` sometimes finishes a proof, sometimes leaves a ring residue” — knowing which it does for a given goal is a matter of experience, but the fix when it surprises you is one line.

Extracting $Z_k(t) = \text{CONST}_k/t^\alpha$ from $t^\alpha \cdot Z_k(t) = \text{CONST}_k$. The naive `field_simp` at `this; linarith` did not close the goal `... = CONST / t^α` because `linarith` could not unify the resulting forms. Cleaner: `rw [eq_div_iff htrpow_ne, mul_comm]; exact hZt`. The `eq_div_iff` lemma is the canonical Mathlib idiom for “ $A = B / C$ iff $A * C = B$ ”; combined with a `mul_comm` to align with the existing $t^\alpha \cdot Z_k(t)$ form, the proof is two rewrites.

`Filter.Eventually` has no `.symm`. The `Tendsto.congr’` step at the end of the unconditional theorem takes an `EventuallyEq` and a `Tendsto`; my `heq_ratio` was stated in the direction “LHS = RHS”, but `Tendsto.congr’` wanted the symmetric direction. The reflex `heq_ratio.symm` produces a type error because `Filter.Eventually` doesn’t have a field projection `.symm` — `EventuallyEq` is a `def`, not a structure. The canonical idiom is `(heq_ratio.mono fun _ h => h.symm)`, which lifts the per-element symmetry through the filter. Worth promoting to the laplace `CLAUDE.md`.

Deliberation flag. GPT-5.5 Pro’s deliberation correctly identified the structural risk in the unconditional form: “*This does **not** mean $J_k \sim 0$ in Mathlib’s *IsEquivalent* sense. The correct zero-case translation is $J_k = o[atTop] Z_k$.*” That zero-case corollary is naturally a ~ 5 -line follow-up if any downstream consumer needs it; deferred from this tide. The voting converged on the first round.

5 What was Mathlib, what was new

Mathlib. Standard tendsto chasing on the constant: the two `Tendsto` combinators `div_const` and `congr’`, the `rpow / Gamma / div_rpow` positivity lemmas, the `eq_div_iff` idiom, and the `IsEquivalent-to-ratio` bridge.³

New. Two theorems and zero infrastructure. Everything else — the J-function asymptotic itself, the closed-form partition, the positivity facts — was already present from N1 and the broader seabed.

The tide stays well-bounded in scope. The 200-line size is structural overhead (positivity, constant-massaging, `Pi.div` cleanup) rather than proof difficulty.

6 Lessons

1. **Closed-form denominators turn “asymptotic-equivalence” into “`Tendsto.div_const`”.** When the denominator has an exact closed form (as opposed to merely an asymptotic), the ratio of two t -dependent quantities becomes a one-step `Tendsto` on the numerator divided by a constant. This is much cleaner than chasing two simultaneous asymptotics. Worth flagging for future tides where the denominator is a partition function: don’t reach for `Tendsto.div` when `Tendsto.div_const` suffices.
2. **The `IsEquivalent` idiom requires eventual non-vanishing of the target.** Mathlib’s bridge to ratio convergence is `isEquivalent_iff_tendsto_one`, which assumes g is eventually nonzero. For paper-style statements $f \sim c \cdot g$ with c a parameter that might be zero, the `IsEquivalent` form requires $c \neq 0$ as a hypothesis; the unconditional ratio form does not. So the natural public API is the pair: unconditional ratio + `IsEquivalent` corollary under non-vanishing of the leading coefficient. Both are useful for different downstream contexts.
3. **`Filter.Eventually` symmetry needs `.mono`, not `.symm`.** The reflex `heq.symm` fails on `EventuallyEq` (it’s a `def`, not a structure). The canonical idiom is `(heq.mono fun _ h => h.symm)` — lift the per-element symmetry through the filter. Worth promoting to the laplace CLAUDE.md under the existing “`Pi.add` vs single-lambda” entry; both are filter-level vs. point-level mismatches.

7 Follow-ups

- **The $\pi(0) = 0$ corollary $J_k = o[atTop] Z_k$.** GPT flagged this as the correct *little-o* form for the zero-case. ~ 5 lines on top of the unconditional ratio theorem (`Tendsto.div_const` of $J_k/Z_k \rightarrow 0$

³Specifically `Real.rpow_pos_of_pos`, `Real.Gamma_pos_of_pos`, `Real.div_rpow`, `eq_div_iff`, and `Asymptotics.isEquivalent_iff_tendsto_one`.

via `IsLittle0`'s `Tendsto`-characterisation). Defer until a downstream consumer asks; cheap to add then.

- **The $k = 2$ and $k = 3$ specialisations.** The grammar paper, the primer, and the L1 retrospective all reference the quartic and sextic $\sim$ -forms specifically. With the parametric form in hand these are zero-line corollaries; named wrappers under `quartic_jfunction_isEquivalent` and the obvious sextic analogue would make the `\leanref` story for those papers clean. ~ 10 lines each. Best landed once the relevant paper-side has TeX statements to tag.
- **Hypothesis weakening (L3 in the v3 survey).** The present tide inherits N1's `Continuous prior + globally bounded hypothesis` class. The minimal hypothesis class for the underlying DCT step is `AEMeasurable + ContinuousAt 0 + global bound`, per the L3 follow-up. Refactoring the parametric J_k asymptotic to the weaker class would automatically propagate to the `_isEquivalent` wrapper here. Bigger tide.
- **Promote the partition closed form to a named `kth_partition_smul_pow` or analogous.** The rearrangement $t^\alpha \cdot Z_k(t) = \text{CONST}_k$ is a useful canonical form for any tide that wants to reason about ratios of J -like integrals. Pulling it out as a named one-line corollary of `kth_partition` would shave ~ 15 lines off the present file's `set + algebra` block, and become reusable. Cheap; defer until a second consumer materialises.